2. Mono Alphabetic Cipher

To implement a **Monoalphabetic Caesar Cipher** in C for encrypting and decrypting a plaintext message using a fixed substitution key.

Algorithm

Encryption Algorithm

1. Start.
2. Read the plaintext message.
3. Read the shift key.
4. For each character in the plaintext:
    - If it is an uppercase letter, shift it forward by the key.
    - If it is a lowercase letter, shift it forward by the key.
    - Leave spaces and special characters unchanged.
5. If the shift goes beyond Z or z, wrap around to the beginning of the alphabet.
6. Display the encrypted text.
7. Stop.

Decryption Algorithm

1. Start.
2. Read the encrypted text and key.
3. For each character:
    - Shift the character backward by the key.
    - If the shift goes before A or a, wrap around to the end of the alphabet.
4. Display the decrypted text.
5. Stop.

Program :

```
#include <stdio.h>

int main() {
    char text[100];
    int key, i;

    printf("Enter the text: ");
```

```
fgets(text, sizeof(text), stdin);

printf("Enter the key: ");
scanf("%d", &key);

key = key % 26;

// Encryption
for (i = 0; text[i] != '\0'; i++) {

    if (text[i] >= 'A' && text[i] <= 'Z') {
        text[i] = (text[i] - 'A' + key) % 26 + 'A';
    }

    else if (text[i] >= 'a' && text[i] <= 'z') {
        text[i] = (text[i] - 'a' + key) % 26 + 'a';
    }
}

printf("\nEncrypted Text: %s", text);

// Decryption
for (i = 0; text[i] != '\0'; i++) {

    if (text[i] >= 'A' && text[i] <= 'Z') {
        text[i] = (text[i] - 'A' - key + 26) % 26 + 'A';
    }

    else if (text[i] >= 'a' && text[i] <= 'z') {
        text[i] = (text[i] - 'a' - key + 26) % 26 + 'a';
    }
}
```

```
    printf("Decrypted Text: %s", text);

    return 0;
}
```

Output